

Yihan Zhang
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EDUCATION

- M.S **Nanyang Technological University, Singapore**
Expected May 2027 | **MSc in Artificial Intelligence in Medicine** - *Lee Kong Chian School of Medicine*
- B.S **The University of North Carolina at Chapel Hill, NC, USA**
Dec 2025 | **Information Science; Minor in Studio Art & Geography**

CORE COURSEWORK

GIS Analysis with R (A)
Information Visualization (A)
Product Management (A)
Human-Computer Interaction UI/UX Design (A)
Object-Oriented Programming with Java
Applied Arts [Painting, Photography] (A)
3D Design Studio (A) – Maya-based 3D animation & rendering
Global French-Speaking World (A) – critical writing & conference-level research

INDUSTRIAL R&D & SELECTED PATENTS

R&D Team Member, Wenzhou Jiawei Environmental Technology Co., Ltd. (Subsidiary of Weiming Group, a Chinese A-Share Listed Company)

Contributed to industrial hardware R&D, resulting in co-authorship on multiple utility model patents. (**Research through Design with Industry background**)

First Inventor, China Utility Model Patent (CN209791046U) * Title: A Disc-Tube Reverse Osmosis Apparatus for Landfill Leachate. * Contribution: Solved a critical **human-centered usability problem** ("difficult to clean") by designing a detachable filter mesh structure , transforming a maintenance bottleneck into a serviceable system.

Co-Second Inventor, China Utility Model Patent (CN215209304U) * Title: An Extraction Device for Kitchen Waste Gutter Oil. * **Contribution:** Co-designed a complex mechanical apparatus for multi-stage pressing, filtering, and waste separation , proving capability in industrial hardware/ Engineer design.

SELECTED WORK

- "Pluggy": Tangible Digital Wellbeing Ecosystem** | *Capstone Winner (1st Place) System Architect & HCI Lead Researcher* | Fall 2025
- **Role:** System Architect & HCI Lead Researcher. Addressed "Revenge Bedtime Procrastination" through a novel hardware-software ecosystem.
 - **System Design:** Architected a "phygital" intervention connecting a **Smart Charger**

(Hardware) with a Gamified App (Software). Designed the "Progress Loss" mechanic where disconnecting the phone prematurely triggers in-game penalties, leveraging Loss Aversion psychology to enforce sleep habits.

- **Research & Prototyping:** Conducted mixed-method user research to define user friction points. Developed the Product Design Rationale and system architecture, bridging physical constraints with digital rewards to promote healthy sleep hygiene.

“Super Ramses”: 3D Character Animation Short | 3D Design Studio (MEJO 585) | 2025 Fall

- Created a narrative 3D animation short in Autodesk Maya, from character blocking to final rendered sequence.
- Implemented camera choreography (dolly, pan, close-up shots) and physically based lighting; used Arnold for rendering and experimented with Bifrost-based snow / particle effects for environmental storytelling.
- Optimized the full pipeline (model → rig → animation → simulation → render) under tight course deadlines, demonstrating ability to manage both artistic direction and technical constraints.

“Chill by U”: Modular Laptop Stand | *Winner, 2nd Most Creative Prize* – UNC MakerFest 2024

- Created a modular stand tailored to college students’ multitasking habits: integrated phone dock, pen holder, and cup mount.
- Added Arduino-controlled fan system based on real-time thermal feedback for efficient laptop cooling.
- Constructed with laser-cut wood and 3D-printed modular joints, enabling reconfiguration for different workspace setups.
- Focused on daily user experience and ergonomic optimization, with attention to aesthetic detailing and portability.

SmartCan: Public Sorting Assistant for Trash Behavior Correction

User-Centered Design | Interaction Design Project

- Designed a UI-integrated trash bin system that visualizes behavioral feedback to encourage proper sorting.
- Applied social incentive models and behavioral economics principles to shape long-term user habits.
- Featured visual display, motion sensor activation, and responsive lighting to increase interaction and awareness in shared spaces.
- Emphasized urban micro-interaction with implications for city-scale behavior intervention systems.

Smart Feeder: Intelligent Bird Feeding System

Arduino-based Autonomous Device | UNC Maker Project

- Developed an intelligent bird feeder using photoresistor and pressure sensors to distinguish birds from intruding animals (e.g., squirrels).
- Integrated automatic gate control based on weight and light input; planned for future upgrade with YOLOv8 image recognition.
- Emphasized sustainability with over 90% recycled materials (e.g., acrylic panels,

reused plastic bottles, biodegradable PLA).

- Addressed real-world need for remote feeding and species-specific delivery via behavior-informed control logic.

Lake Surface Cleaning Robot | Vessel Environmental Robotics Concept

Environmental Robotics Concept Solution Design

- Developed a robot system to clean surface trash in lakes.
- Designed a double-layered collection net structure with floating sensors to control water-level adaptation.
- Used Arduino motor control and real-time feedback mechanism.
- Emphasized environmental sensing and hardware feasibility in open water environments.

Population Change and Spatial Autocorrelation in the U.S. (2020–2024)

Geospatial Analysis with R | R - Data & GIS Analysis Project

- Used *R*, *sf*, *spdep*, and *tmap* packages to analyze population growth patterns and spatial autocorrelation across U.S. states.
- Built spatial weight matrices, calculated Moran's *I*, and visualized hot spots.
- Integrated socioeconomic data (unemployment rates) with Census and shapefiles for deeper interpretation.
- Delivered technical and policy-relevant insights on demographic shifts and clustering.

ArcGIS Story Map Blog: Spatial Inequality in the Global Coffee Trade

Global Development & Inequality Analysis with Data Analysis

- Created an interactive ArcGIS Story Map comparing coffee trade systems in Vietnam and Ethiopia.
- Explored how spatial factors like infrastructure and land access shape economic outcomes.
- Incorporated visual data layers, institutional actor mapping, and Gini coefficient overlays.
- Synthesized development theory (colonial geographies, capability theory) with visual storytelling.

PROFESSIONAL EXPERIENCE

China Construction Bank, Institutional Business Department | *Intern*

Product & Information Management

Zhejiang, China | May 2024 – Aug 2024

- Participated in the development of consumer-oriented digital products for banking applications.
- Assisted in the design of a Human Resources Information Management System (HRIMS) for a major institutional client.
- Contributed to full-stack development: grid architecture, coding, bug fixing, and documentation.
- Engaged in team-based technical discussions and iterative testing.

Weiming Environmental Protection Co., Ltd. | *IT Operations Intern*

Zhejiang, China | Jun 2022 – Aug 2022

- Assisted in web-based development for industrial applications, including front-end functionality and backend API integration.
- Developed Python + Selenium automation scripts to monitor system status; reduced system failure response time and decreased maintenance ticket volume by 40%.
- Compared EU and Chinese environmental regulations to incorporate relevant compliance features into platform design.

HONORS AND AWARDS

2025 Selected Panelist — **30th Carolina Conference for Romance Studies**
(Research Panel)

Invited to present the research paper "Humble Before the Storm" based on academic excellence in French Studies; recognized for critical analysis of Indigenous environmental philosophy.

2024 UNC MakerFest — 2nd Most Creative Prize (“**Chill by U**” Laptop Stand)

2022 Let's Charrette & Sustainable Campus Design Competition, 3rd Campus topic Prize

2025 Selected Exhibitor — UNC Smart Design Showcase,

2024 UNC Academic Excellence Dean's List (3 Times)

TECHNICAL SKILLS

Programming: Python, Java, R, JavaScript, HTML5, Matlab

Tools: Adobe Illustrator, Figma, Arduino, ArcGIS, Rhino, Unity, SQL, After Effects

Prototyping: Laser cutting, 3D printing, soldering, 3DMax, TinkerCAD, Vinyl cutter

Others: Git, LaTeX, Blender, Photoshop, LightRoom, PR

LANGUAGES

Mandarin (Native), English (Professional Efficient), Japanese (Beginner)